Lecture 8: Expectation-Maximization
Gaussian Mixture Models, Step by Step

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1. EM Algorithm
   - Expectation-Maximization Algorithm

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Motivation of EM: the labels are the missing piece

For a GMM, $p(\mathbf{x}_n, z_n = k \mid \boldsymbol{\theta}) = \pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.

$$\ell(\boldsymbol{\theta}) = \sum_n \log \sum_k \pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

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EM: infer probabilities with the current model, then refit using those probabilities.

Notation: an index is different from an indicator

For $K = 3$, suppose input n came from component 2.

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$q_{kn} = p(z_n = k \mid \mathbf{x}_n, \boldsymbol{\theta})$ is a probability, not a component index.

$$\mathbb{E}[z_{nk} \mid \mathbf{x}_n, \boldsymbol{\theta}] = q_{kn}.$$

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Then recompute responsibilities. Old probabilities need not fit the new model.

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Normalize across the possible components for this input:

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Check: $q_{kn}^{(t)} \geq 0$ and $\sum_k q_{kn}^{(t)} = 1$ for every n.

Bridge: E-step on the same three inputs

Use $x = (0, 2, 4)$, $\mu_1 = 0$, $\mu_2 = 4$, $\sigma_1^2 = \sigma_2^2 = 4$, and $\pi_1 = \pi_2 = 1/2$.

x_n	0	2	4

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The middle input shares its weight equally. The endpoints remain uncertain.

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$$\sum_k N_k^{(t)} = \sum_n \underbrace{\sum_k q_{kn}^{(t)}}_{1} = N.$$

Each input contributes a total weight of one, distributed across components.

The M-step: weighted center and mixture proportion

$$\boldsymbol{\mu}_k^{(t+1)} = \frac{\sum_n q_{kn}^{(t)} \mathbf{x}_n}{N_k^{(t)}}, \qquad \pi_k^{(t+1)} = \frac{N_k^{(t)}}{N}.$$

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For component 1 in the same example:

$$\mu_1^{\text{new}} \approx \frac{0.8808 \cdot 0 + 0.5000 \cdot 2 + 0.1192 \cdot 4}{1.500} \approx 0.985.$$

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K-means uses $0/1$ counts; EM uses posterior weights in the same averaging operation.

The M-step: spread around the new center

Freeze the same $q_{kn}^{(t)}$, and use the **new** mean:

$$\boldsymbol{\Sigma}_k^{(t+1)} = \frac{\sum_n q_{kn}^{(t)} (\mathbf{x}_n - \boldsymbol{\mu}_k^{(t+1)})(\mathbf{x}_n - \boldsymbol{\mu}_k^{(t+1)})^\top}{N_k^{(t)}}.$$

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In one dimension, this is just a weighted average of squared residuals:

$$(\sigma_1^2)^{\text{new}} \approx \frac{0.8808(0 - 0.985)^2 + 0.5(2 - 0.985)^2 + 0.1192(4 - 0.985)^2}{1.5} \approx 1.636.$$

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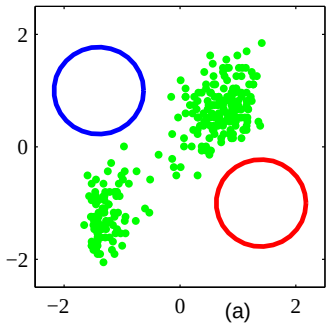
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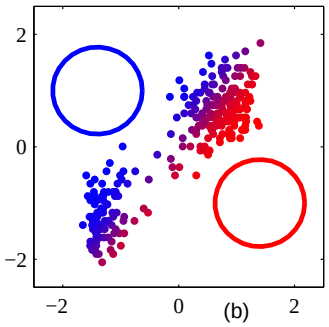
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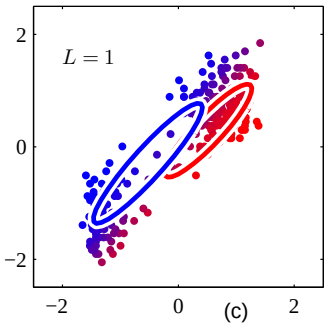
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Symmetry gives $(\sigma_2^2)^{\text{new}} \approx 1.636$.
Now repeat the E-step with the new centers, variances, and proportions.

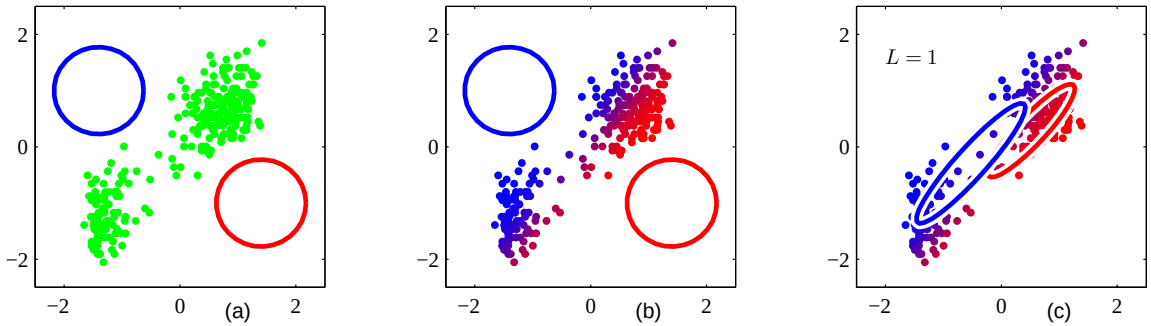
EM for GMM: watch assignment and shape change





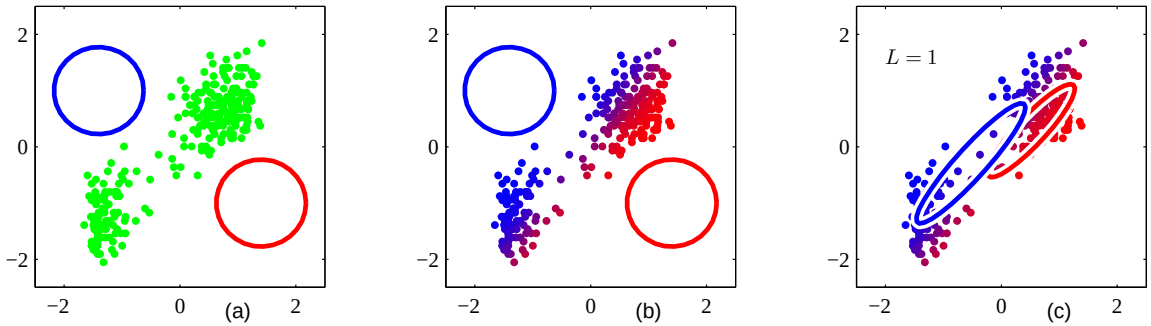


EM for GMM: watch assignment and shape change



Left: initial Gaussians. Middle: infer memberships. Right: refit the Gaussians.

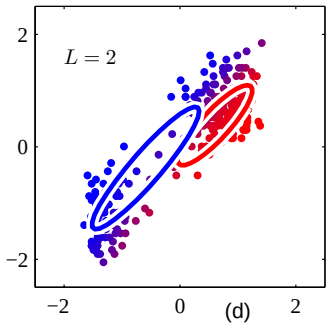
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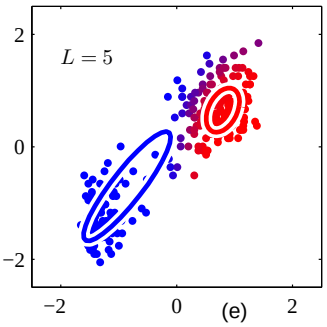


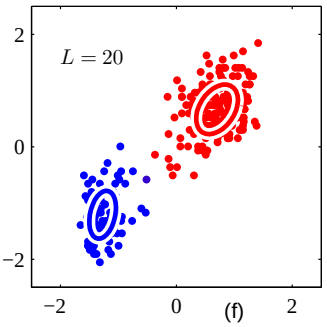
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The parameters move because each input contributes in proportion to its responsibility.

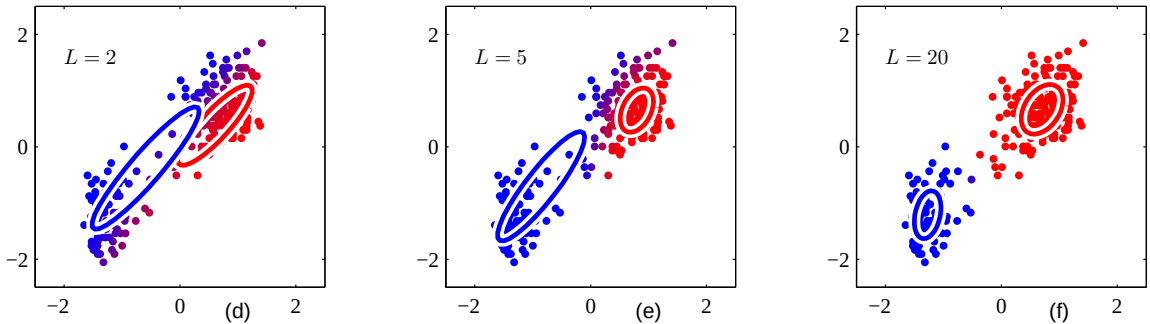
EM for GMM: repeat with the updated model





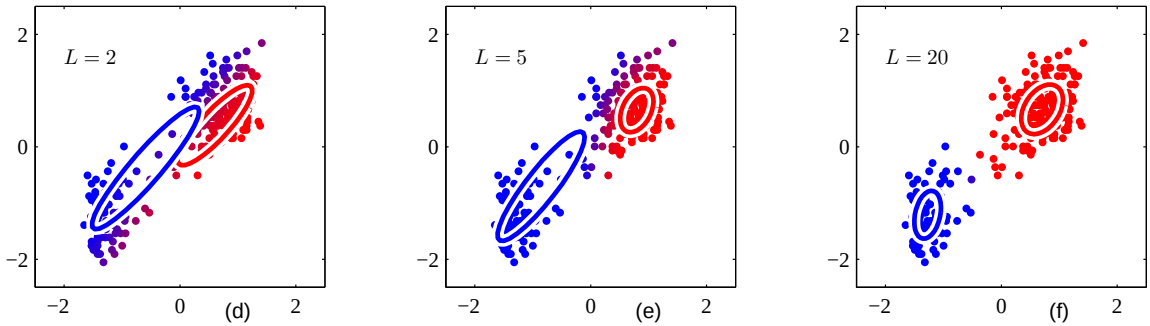


EM for GMM: repeat with the updated model



New parameters change the responsibilities; new responsibilities change the fit.

EM for GMM: repeat with the updated model



New parameters change the responsibilities; new responsibilities change the fit.

We can now run EM. Next: what objective makes these updates sensible?

M-step objective: average over the missing labels

With known labels, the complete-data log-likelihood is

$$\log p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = \sum_{n,k} z_{nk} [\log \pi_k + \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)].$$

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Average this expression using the **old model's** posterior:

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(t)}) = \sum_{n,k} q_{kn}^{(t)} [\log \pi_k + \log \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)].$$

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The M-step maximizes Q over $\boldsymbol{\theta}$ while keeping those weights fixed.

M-step Derivation: the weighted mean

Retain only the mean-dependent terms of Q:

$$Q_k(\boldsymbol{\mu}_k) = -\frac{1}{2}\sum_n q_{kn}^{(t)}(\mathbf{x}_n - \boldsymbol{\mu}_k)^\top \boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_n - \boldsymbol{\mu}_k) + \text{const.}$$

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Differentiate, set to zero, and multiply by $\boldsymbol{\Sigma}_k$:

$$\sum_n q_{kn}^{(t)} \boldsymbol{\Sigma}_k^{-1}(\mathbf{x}_n - \boldsymbol{\mu}_k) = 0 \implies \sum_n q_{kn}^{(t)} \mathbf{x}_n = N_k^{(t)} \boldsymbol{\mu}_k.$$

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$$\boldsymbol{\mu}_k^{(t+1)} = \frac{\sum_n q_{kn}^{(t)} \mathbf{x}_n}{N_k^{(t)}}.$$

Same stationary equation as K-means, with fractional weights.

M-step Derivation: the weighted covariance

Use the new mean. Write $S_k = \sum_n q_{kn}^{(t)} (\mathbf{x}_n - \boldsymbol{\mu}_k^{(t+1)})(\mathbf{x}_n - \boldsymbol{\mu}_k^{(t+1)})^\top$.

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Use $\nabla_{\boldsymbol{\Lambda}} \log |\boldsymbol{\Lambda}| = \boldsymbol{\Lambda}^{-1}$ for symmetric positive definite $\boldsymbol{\Lambda}$:

$$\frac{N_k^{(t)}}{2} \boldsymbol{\Lambda}_k^{-1} - \frac{1}{2} S_k = 0 \quad \Longrightarrow \quad \boldsymbol{\Sigma}_k^{(t+1)} = \frac{S_k}{N_k^{(t)}}.$$

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If S_k is singular, there is no positive definite interior solution.

M-step Derivation: the mixture proportions

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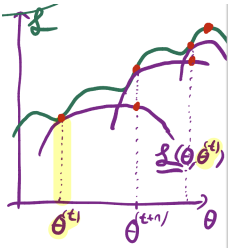
Sum over k: $N = \lambda$, because $\sum_k N_k^{(t)} = N$.

$$\pi_k^{(t+1)} = \frac{N_k^{(t)}}{N}.$$

A component's mixture proportion is its share of the total effective count.

Why can maximizing Q help the observed likelihood?

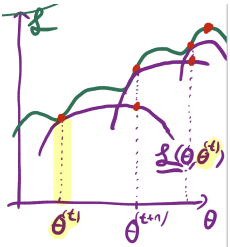
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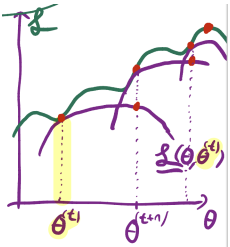
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Raise this touching bound $\Rightarrow$ the true likelihood cannot decrease.
We will build the bound one input at a time.

Build the lower bound: insert weights, then use log concavity

For one input, let $a_k(\boldsymbol{\theta}) = \pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. Choose $q_k > 0$ with $\sum_k q_k = 1$.

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This is Jensen's inequality: $\log(\text{weighted average}) \geq$ weighted average of logs.

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Equality holds when all ratios $a_k(\boldsymbol{\theta})/q_k$ are equal.

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The posterior probabilities we already computed make the bound touch the likelihood.

The Evidence Lower Bound: expected fit plus entropy

Sum the per-input bounds. Write $q_{kn} = q_n(z_n = k)$:

$$\mathcal{F}(q, \boldsymbol{\theta}) = \sum_{n,k} q_{kn} \log \frac{\pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{q_{kn}} \leq \ell(\boldsymbol{\theta}).$$

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$$\mathcal{F}(q, \boldsymbol{\theta}) = \underbrace{\sum_{n,k} q_{kn} \log[\pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)]}_{\text{expected complete-data log-likelihood}} + \underbrace{\left(- \sum_{n,k} q_{kn} \log q_{kn} \right)}_{H(q)\text{: entropy}}.$$

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In the M-step, $q = q^{(t)}$ is fixed, so $H(q^{(t)})$ is constant.
Maximizing $\mathcal{F}$ over $\boldsymbol{\theta}$ is exactly maximizing Q.

EM algorithm: the three-line improvement argument

E-step: $q^{(t)}$ is the posterior under $\boldsymbol{\theta}^{(t)}$.

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Touch the old value $\rightarrow$ raise the bound $\rightarrow$ reach a likelihood no smaller.

Background: KL measures the gap between distributions

For discrete probabilities on a common support:

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In EM, compare our distribution over hidden labels with the model's posterior.

ELBO: the exact gap is a posterior KL

For $q(\mathbf{Z}) = \prod_n q_n(z_n)$, use $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}) p(\mathbf{X} \mid \boldsymbol{\theta})$:

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&= \ell(\boldsymbol{\theta}) - D_{\mathrm{KL}} \left(q(\mathbf{Z}) \| p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}) \right).
\end{aligned}$$

E-step: zero gap at the old model. M-step: raise $\mathcal{F}$ with q fixed.

The relationship to K-means: hard and soft assignments

	K-means	GMM with EM
Assignment	$z_{nk} \in \{0, 1\}$	$q_{kn} \in [0, 1]$

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Geometry	Euclidean distance	learned covariance

Both alternate assignment and fitting; their objectives are different.

When does a soft assignment become a nearest-center choice?

Constrain all components to a common fixed $\boldsymbol{\Sigma}_k = \sigma^2 \mathbf{I}$.

$$q_{kn} = \frac{\pi_k \exp[-\|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2/(2\sigma^2)]}{\sum_j \pi_j \exp[-\|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2/(2\sigma^2)]}.$$

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For the unique nearest center k_*, and any other j:

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This uses common *isotropic* covariance and positive fixed proportions.
A diagonal covariance alone is not sufficient.

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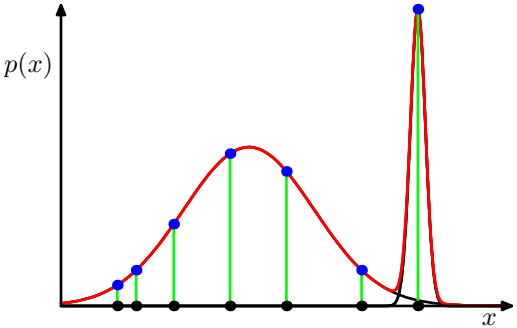
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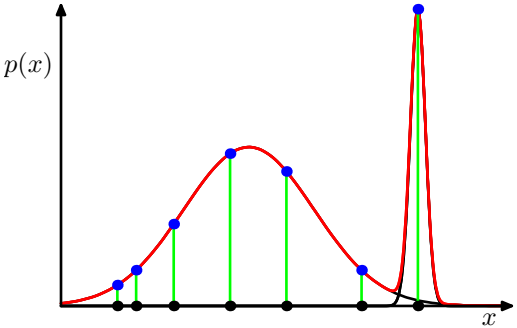
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- This alone does not guarantee a local maximum or parameter convergence.
- General GMM likelihoods can be unbounded: a component can collapse onto one input.

Inspect covariances and effective counts as well as the likelihood.

Diagnosis: why an ever-rising likelihood can be a warning

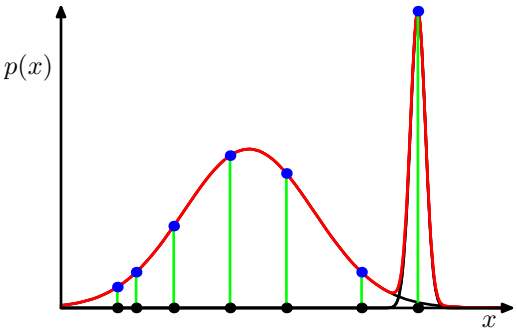


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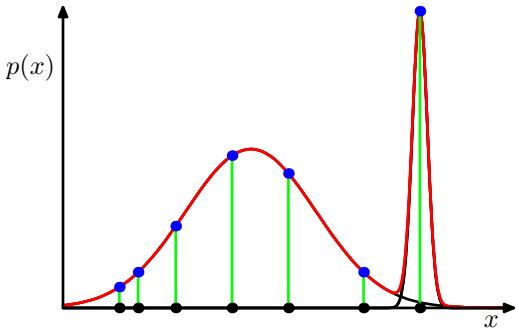
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Other components can still explain the remaining inputs. A larger likelihood may mean collapse.

Use covariance constraints or regularization, and monitor tiny N_k.

Practical Considerations: connect each fix to a failure

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- **Underflow in tiny densities:** normalize log weights using log-sum-exp.
- **Choosing K:** compare held-out fit; AIC/BIC are useful heuristics, with care for mixture singularities.

Summary of EM for GMM: one complete iteration

E: freeze $\theta^{(t)}$. For every input, normalize component weights:

$$q_{kn}^{(t)} \propto \pi_k^{(t)} \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k^{(t)}, \boldsymbol{\Sigma}_k^{(t)}), \quad \sum_k q_{kn}^{(t)} = 1.$$

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Evaluate $\ell(\boldsymbol{\theta}^{(t+1)})$, check for degeneracy, then repeat.
Why it improves: the touching lower bound. Why it can fail: covariance collapse.

Applications of EM: identify what is missing

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E: infer a distribution over what is missing.
M: maximize the expected complete-data log-likelihood.

Optional extension: EM and Variational Inference

The same identity guides both methods:

$$\ell(\boldsymbol{\theta}) = \mathcal{F}(q, \boldsymbol{\theta}) + D_{\mathrm{KL}}(q \| p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta})).$$

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A restricted family may leave a positive gap; improving its ELBO need not improve the true likelihood.

Table of Contents

2 Optional: Bregman connections

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The quadratic term is a Bregman divergence. The full negative log density also contains a normalization term.

Return to the EM summary

The Unifying Tool: Bregman Divergence

Let F be differentiable and strictly convex on an open convex domain.

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Convexity puts the tangent below F, so $D_F \geq 0$.
Strict convexity gives $D_F = 0$ only when $\mathbf{p} = \mathbf{q}$.

The Unifying Tool: Bregman Divergence

Let F be differentiable and strictly convex on an open convex domain.

$$D_F(\mathbf{p}, \mathbf{q}) = F(\mathbf{p}) - F(\mathbf{q}) - \nabla F(\mathbf{q})^\top (\mathbf{p} - \mathbf{q}).$$

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It is a gap above a tangent approximation. It need not be symmetric or satisfy the triangle inequality.

Connection 1: squared Euclidean distance

Take $F(\mathbf{x}) = \mathbf{x}^\top \mathbf{x}$, so $\nabla F(\mathbf{x}) = 2\mathbf{x}$.

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$$\begin{aligned} D_F(\mathbf{x}, \boldsymbol{\mu}) &= \mathbf{x}^\top \mathbf{x} - \boldsymbol{\mu}^\top \boldsymbol{\mu} - 2\boldsymbol{\mu}^\top (\mathbf{x} - \boldsymbol{\mu}) \\ &= \mathbf{x}^\top \mathbf{x} - 2\boldsymbol{\mu}^\top \mathbf{x} + \boldsymbol{\mu}^\top \boldsymbol{\mu} = \|\mathbf{x} - \boldsymbol{\mu}\|^2. \end{aligned}$$

K-means minimizes the sum of these divergences to the assigned centers.

Connection 2: a Gaussian with fixed covariance

Fix one positive definite covariance $\boldsymbol{\Sigma}$ for all components.

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For this Gaussian family:

$$-\log \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = D_F(\mathbf{x}, \boldsymbol{\mu}) + \underbrace{\tfrac{D}{2}\log(2\pi) + \tfrac{1}{2}\log|\boldsymbol{\Sigma}|}_{\text{constant when covariance is fixed}} .$$

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When covariances are learned separately, the log-determinant varies across components and must be retained.

Hard versus soft: keep the assumptions visible

For the fixed common covariance case, let $d_{kn} = D_F(\mathbf{x}_n, \boldsymbol{\mu}_k)$.

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For $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, see the small-variance limit.
Further reading: Banerjee et al., Clustering with Bregman Divergences (2005).
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